

CASE 3

ln_cpi ln_cif ln_customs_duties_taxes ln_opex ir ln_fx
m2 m3 m4 m5 m6 m7 m8 m9 m10 m11 m12

Akaike Info Criterion (AIC)

HAC (Newey Test)

Bounds Tests

Null hypothesis: No levels relationship
Number of dynamic cointegrating variables: 5
Deterministics: Unrest. constant (Case 3)
Sample size: 176

Test Statistic	Value
F-statistic	2.234561
t-statistic	-1.491498

Bounds Critical Values

Sample Size		10%	5%	1%
F-Statistic				
I(0)	Asymptotic	2.260	2.620	3.410
I(1)	Asymptotic	3.350	3.790	4.680
t-Statistic				
I(0)	Asymptotic	-2.570	-2.860	-3.430
I(1)	Asymptotic	-3.860	-4.190	-4.790

* I(0) and I(1) are respectively the stationary and non-stationary bounds.

Conditional Error Correction

Dependent Variable: LNCPI

Method: ARDL

Date: 09/08/26 Time: 22:48

Sample: 2011M05 2025M12

Included observations: 176

Lag selection: Automatic (deplags=4, reglags=4)

Selected model: ARDL(4,0,0,1,1,0) using AIC (12500 models evaluated)

HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
Long-run Regressors				
Linear: Dependent				
LNCPI(-1)	-0.011316	0.007587	-1.491498	0.1379
Linear: Independent				
LN_CIF	-0.000328	0.000215	-1.525561	0.1292
LN_CUSTOMS_DUTIES_TAX...	0.000255	0.000446	0.572674	0.5677
LN_OPEX(-1)	-0.000403	0.000135	-2.975182	0.0034
IR(-1)	5.54E-05	8.31E-05	0.665989	0.5064
LN_FX	0.002203	0.001244	1.771661	0.0784
Short-run Regressors				

Linear: Dependent				
D(LNCPI(-1))	-0.226721	0.066515	-3.408567	0.0008
D(LNCPI(-2))	-0.013086	0.077776	-0.168253	0.8666
D(LNCPI(-3))	0.163875	0.092427	1.773015	0.0782
Linear: Independent				
D(LN_OPEX)	0.000130	0.000250	0.518784	0.6047
D(IR)	-0.000336	0.000219	-1.531351	0.1277
Deterministic				
C	0.049044	0.033824	1.449984	0.1491
Fixed				
M2	-0.000506	0.000316	-1.603137	0.1110
M3	-0.000413	0.000252	-1.639380	0.1032
M4	-0.000591	0.000254	-2.328218	0.0212
M5	-0.000488	0.000246	-1.980510	0.0494
M6	-0.000573	0.000289	-1.982305	0.0492
M7	-3.28E-05	0.000317	-0.103605	0.9176
M8	-0.000484	0.000238	-2.036985	0.0434
M9	-0.000493	0.000290	-1.697936	0.0916
M10	-0.000351	0.000271	-1.293444	0.1978
M11	-0.000388	0.000283	-1.369781	0.1728
M12	-0.000424	0.000233	-1.819971	0.0707

R-squared	0.217533	Mean dependent var	0.000321
Adjusted R-squared	0.105021	S.D. dependent var	0.000684
S.E. of regression	0.000647	Akaike info criterion	-11.72614
Sum squared resid	6.41E-05	Schwarz criterion	-11.31181
Log likelihood	1054.900	Hannan-Quinn criter.	-11.55809
F-statistic	1.933425	Durbin-Watson stat	1.993957
Prob(F-statistic)	0.011094		

Dependent Variable: LNCPI

Method: ARDL

Date: 09/08/26 Time: 22:48

Sample: 2011M05 2025M12

Included observations: 176

Lag selection: Automatic (deplags=4, reglags=4)

Selected model: ARDL(4,0,0,1,1,0) using AIC (12500 models evaluated)

HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
Cointegrating Equation				
COINTEQ	-0.011316	0.003273	-3.456816	0.0007
Short-run Regressors				
Linear: Dependent				
D(LNCPI(-1))	-0.226721	0.060338	-3.757517	0.0002
D(LNCPI(-2))	-0.013086	0.069425	-0.188492	0.8507
D(LNCPI(-3))	0.163875	0.085117	1.925295	0.0560
Linear: Independent				
D(LN_OPEX)	0.000130	0.000230	0.563462	0.5739
D(IR)	-0.000336	0.000221	-1.520533	0.1304
Deterministic				
C	0.049044	0.013979	3.508503	0.0006

Fixed				
M2	-0.000506	0.000298	-1.696876	0.0917
M3	-0.000413	0.000241	-1.712901	0.0887
M4	-0.000591	0.000246	-2.402081	0.0175
M5	-0.000488	0.000230	-2.117807	0.0358
M6	-0.000573	0.000276	-2.076643	0.0395
M7	-3.28E-05	0.000305	-0.107757	0.9143
M8	-0.000484	0.000222	-2.178916	0.0308
M9	-0.000493	0.000274	-1.801222	0.0736
M10	-0.000351	0.000262	-1.338701	0.1826
M11	-0.000388	0.000285	-1.361289	0.1754
M12	-0.000424	0.000226	-1.873838	0.0628
R-squared	0.217533	Mean dependent var	0.000321	
Adjusted R-squared	0.133343	S.D. dependent var	0.000684	
S.E. of regression	0.000637	Akaike info criterion	-11.78296	
Sum squared resid	6.41E-05	Schwarz criterion	-11.45870	
Log likelihood	1054.900	Hannan-Quinn criter.	-11.65144	
F-statistic	2.583847	Durbin-Watson stat	1.993957	
Prob(F-statistic)	0.001084			

☐ Cointegrating Specification

Deterministics: Unrest. constant (Case 3)

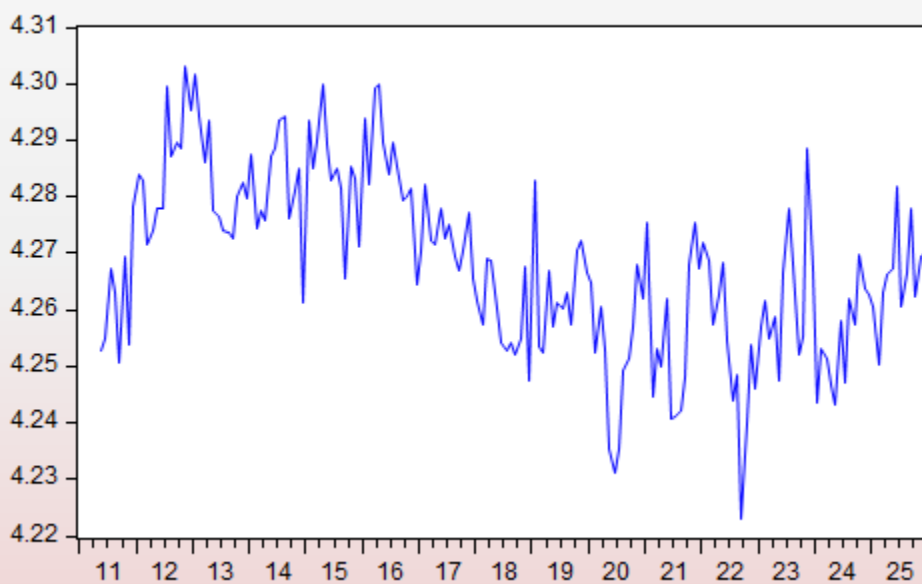
$$\begin{aligned} \text{CE} = & \text{LNCPI}(-1) - (-0.028990 * \text{LN_CIF} + 0.022552 \\ & * \text{LN_CUSTOMS_DUTIES_TAXES} - 0.035600 \\ & * \text{LN_OPEX}(-1) + 0.004892 * \text{IR}(-1) + 0.194699 * \text{LN_FX}) \end{aligned}$$

☐ Cointegrating Coefficients

Deterministics: Unrest. constant (Case 3)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_CIF	-0.028990	0.020244	-1.432070	0.1539
LN_CUSTOMS_DUTIES_TAX...	0.022552	0.031871	0.707612	0.4801
LN_OPEX(-1)	-0.035600	0.022186	-1.604654	0.1104
IR(-1)	0.004892	0.007600	0.643672	0.5207
LN_FX	0.194699	0.166954	1.166183	0.2452

ARDL Cointegrating Series



Ramsey RESET Test

Equation: UNTITLED

Omitted Variables: Squares of fitted values

Specification: LNCPI LNCPI(-1) LNCPI(-2) LNCPI(-3) LNCPI(-4) LN_CIF

LN_CUSTOMS_DUTIES_TAXES LN_OPEX LN_OPEX(-1) IR IR(-1)

LN_FX C M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

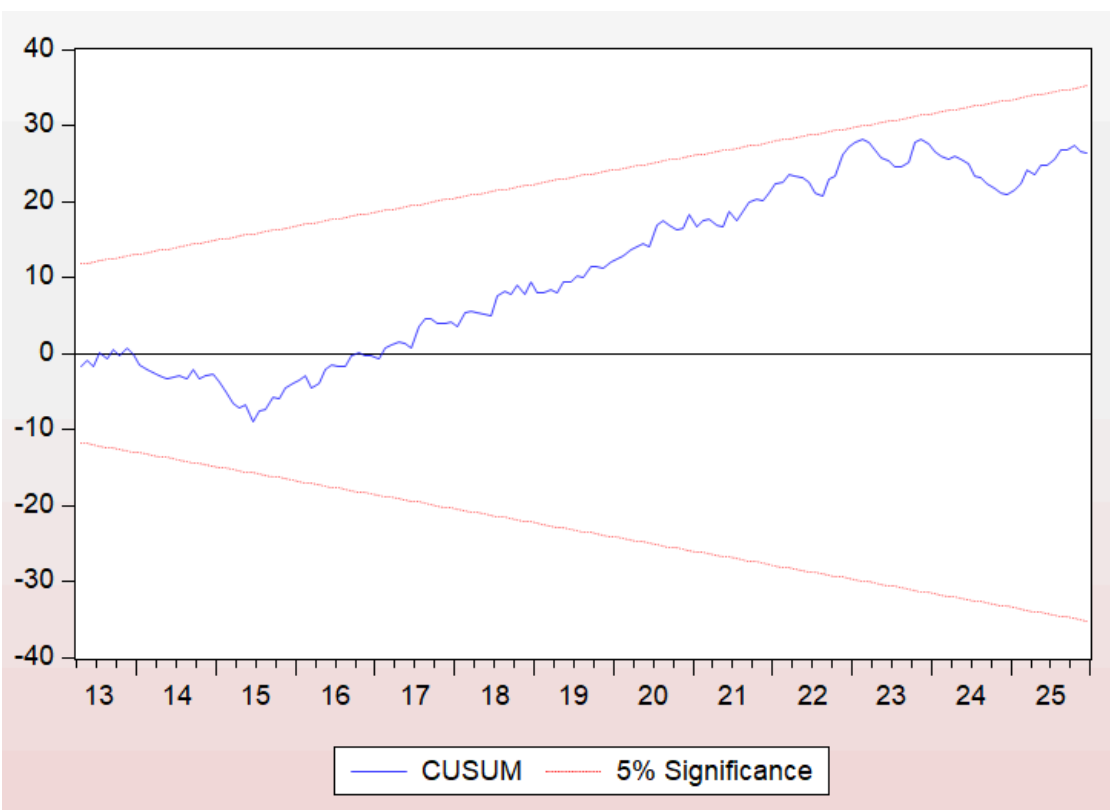
	Value	df	Probability
t-statistic	1.037916	152	0.3010
F-statistic	1.077270	(1, 152)	0.3010
Likelihood ratio	1.242966	1	0.2649

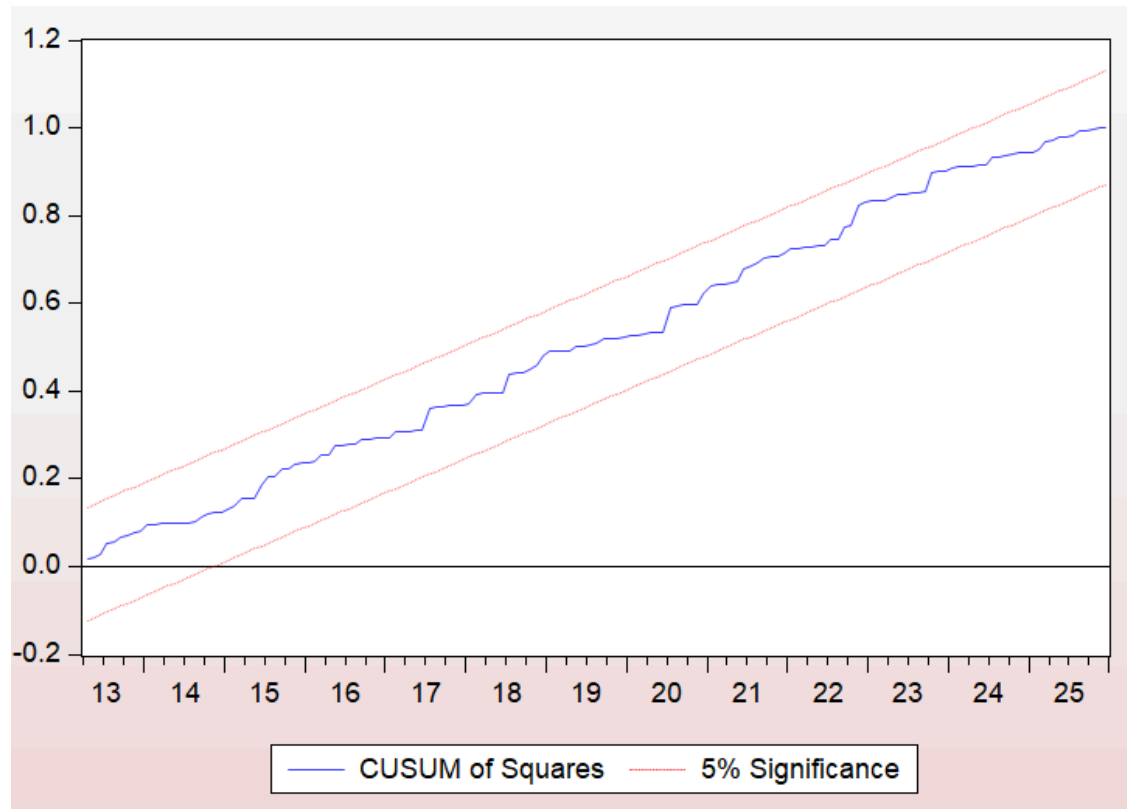
F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	4.51E-07	1	4.51E-07
Restricted SSR	6.41E-05	153	4.19E-07
Unrestricted SSR	6.37E-05	152	4.19E-07

LR test summary:

	Value
Restricted LogL	1054.900
Unrestricted LogL	1055.522



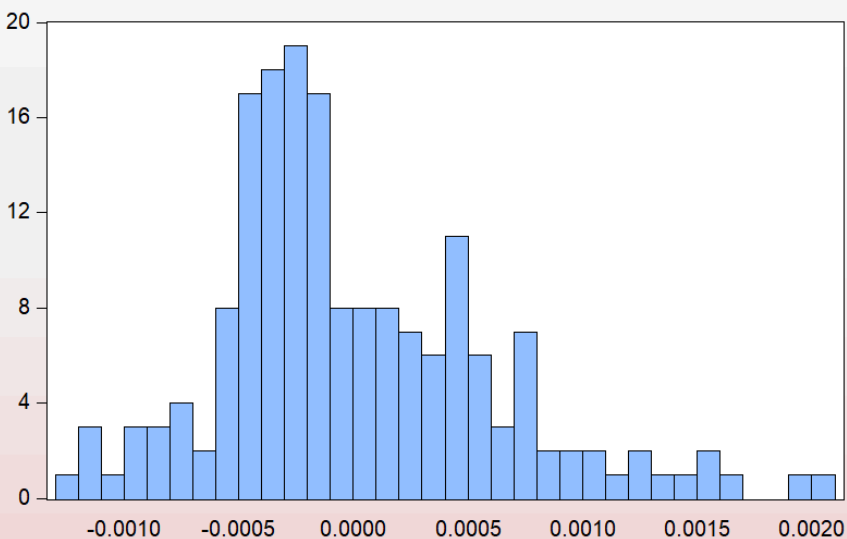


Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	1.262995	Prob. F(22,153)	0.2053
Obs*R-squared	27.05032	Prob. Chi-Square(22)	0.2093
Scaled explained SS	29.22742	Prob. Chi-Square(22)	0.1385

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 2 lags

F-statistic	0.429689	Prob. F(2,151)	0.6515
Obs*R-squared	0.995991	Prob. Chi-Square(2)	0.6077



Series: Residuals
Sample 2011M05 2025M12
Observations 176

Mean 3.55e-16
Median -0.000158
Maximum 0.002066
Minimum -0.001223
Std. Dev. 0.000605
Skewness 0.821755
Kurtosis 3.859502

Jarque-Bera 25.22573
Probability 0.000003

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